COMPUTATIONAL SUPPLEMENT TO "BEYOND LIU'S 0.382709 THRESHOLD FOR THE UNION-CLOSED SETS CONJECTURE"

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Abstract. This supplement documents the finite computational verifications used in the accompanying paper. It identifies the ancillary sources and outputs, specifies the outward-arithmetic model for the interval certificates, records the complete subdivision summaries, and documents the exact rational Bernstein calculation. The mathematical role of each finite assertion is stated in Appendix A of the paper.

Ancillary files and arithmetic model

The ancillary package contains six proposition-specific verification sources and their six output files.

file	verifies	use in the paper
`certify_prop_A1_mpfr.cpp` `certify_prop_A1_mpfr_output.txt`	two-construction pointwise inequality	Proposition A.1; Lemma 5.6; Theorem 1.1
`verify_prop_A2_bernstein.py` `verify_prop_A2_bernstein_output.txt`	exact Bernstein coefficients	Proposition A.2; Lemma 4.3
`certify_prop_A3_mpfr.cpp` `certify_prop_A3_mpfr_output.txt`	Example 4 product lower bound	Proposition A.3; Lemma 6.1
`certify_prop_A4_mpfr.cpp` `certify_prop_A4_mpfr_output.txt`	scalar checks for the analytic regions	Proposition A.4; Lemma 6.3
`certify_prop_A5_mpfr.cpp` `certify_prop_A5_mpfr_output.txt`	local Hessian and stationary point	Proposition A.5; Lemma 6.3
`certify_prop_A6_mpfr.cpp` `certify_prop_A6_mpfr_output.txt`	boundary check and finite cover of the remaining compact region	Proposition A.6; Lemma 6.3; Theorem 1.2

The four C++ drivers for Propositions A.3–A.6 use the shared header `union_closed_interval_common.hpp`; Proposition A.1 is self-contained. The header contains the common interval-arithmetic code and is not a separate certificate.

The C++ code uses outward binary64 interval arithmetic, with MPFR at 192-bit precision for log, exp, and $\sqrt{\ }$. Decimal endpoints are rounded outward before subdivision, midpoint radii are rounded upward, and reported numerical bounds are rounded outward before printing. These conventions prevent gaps caused by floating-point rounding.

The primary build route uses the official MPFR and GMP development headers:

```
for A in A1 A3 A4 A5 A6; do
  g++ -O2 -std=c++17 -frounding-math -fno-fast-math -ffp-contract=off \\
      certify_prop_${A}_mpfr.cpp -lmpfr -lgmp -o certify_${A}
  ./certify_${A}
done
python3 verify_prop_A2_bernstein.py
```

The code is compiled with `-fno-fast-math` and `-ffp-contract=off`; `-frounding-math` is included conservatively. The supplied runs used MPFR version 4.2.2.

1. Two-construction pointwise inequality

This section documents Proposition A.1 using the ancillary source `certify_prop_A1_mpfr.cpp`; the supplied run is recorded in `certify_prop_A1_mpfr_output.txt`.

The two-construction source recomputes the exact constants from the formulas in the paper. It first brackets the unique root x_* of

$$x^4 - 2x^3 + 3x^2 - 1 = 0$$

by opposite directed signs and checks positivity of the derivative on the bracket. It then computes directed intervals for m_* and β. The reported intervals are

$$x_* \in [0.69078759392498001,\ 0.690787593925],$$

$$m_* \in [0.61729091208117659,\ 0.61729091208135323],$$

$$\beta \in [0.10005255986195014,\ 0.10005255986384369].$$

In addition, a separate 256-bit directed MPFR check certifies the displayed decimal bound

$$c_{\mathrm{L}} > 0.382709087918735.$$

It evaluates the polynomial with directed rounding at a certified point below x_*, verifies that $c_{\mathrm{L}}(x)$ is increasing throughout the root bracket, and obtains a directed lower bound above the displayed decimal. It also derives

$$t_* = -\log x_* \in [0.36992289186769,\ 0.36992289186773]$$

and checks that this directed interval lies inside the local rectangle used for strict convexity.

The four scalar checks associated with the analytic regions give

quantity	directed lower bound
$\eta(0.002)$	0.016672474402352748
Lemma 5.4(a) endpoint	0.0011788376979667077
Lemma 5.4(b) endpoint	0.035598095197518845
large-t tail at $t = 4.1$	0.010717682423498683

On the local 40×40 grid, the source computes interval bounds for all entries of the Hessian of the function F defined in Section 5 of the paper. Its global reported bounds are

$$\begin{aligned} \min F_{tt} &= 0.13459156779665421, \\ \min F_{dd} &= 0.87181314699157286, \\ \max |F_{td}| &= 0.4645096021471406, \\ \min(F_{tt}F_{dd} - F_{td}^2) &= 0.050114365218749776. \end{aligned}$$

The determinant bound is formed outward on every individual grid cell rather than inferred from rounded printed extrema.

For the compact complement, set $\xi = d/t$ and $v = t(1-\xi)$. A terminal box is accepted either because it is contained in the local strict-convexity rectangle or because the centered mean-value lower bound is positive. The complete adaptive summary is

t range	$v_{\min}$	terminal	local	splits	depth	least lower bound
$[0.002, 0.1]$	$5 \cdot 10^{-5}$	238	0	110	6	$1.4683343626197839 \cdot 10^{-4}$
$[0.1, 0.34]$	$5 \cdot 10^{-5}$	363	10	213	6	$1.2396204165999228 \cdot 10^{-6}$
$[0.34, 0.40]$	$5 \cdot 10^{-5}$	245	27	144	6	$5.6012230503014744 \cdot 10^{-4}$
$[0.40, 1.4]$	$5 \cdot 10^{-5}$	722	10	540	8	$6.9620924720995344 \cdot 10^{-6}$
$[1.4, 4.1]$	0.1	671	0	351	2	$4.4688168746738737 \cdot 10^{-3}$

There are no unresolved boxes.

2. Exact Bernstein coefficient check

This section documents Proposition A.2 using the ancillary source `verify_prop_A2_bernstein.py`; the supplied run is recorded in `verify_prop_A2_bernstein_output.txt`.

The checker uses only Python's standard-library `fractions.Fraction` class, so all polynomial coefficients, the rescaling by 43/64, and the power-to-Bernstein conversion are evaluated exactly as rational numbers.

Proposition A.2 records the coefficient matrix used in Lemma 4.3 of the paper. The ancillary script `verify_prop_A2_bernstein.py` constructs

$$W = 2 - x - y + xy, \qquad U = 2 - 2x - y + 2xy, \qquad V = 2 - x - 2y + 2xy,$$

and

$$r = xyW, \qquad P(x, y) := WUV - (1 - r)$$

with exact rational coefficients. It then substitutes

$$x = \frac{43}{64}s, \qquad y = \frac{43}{64}t$$

and converts the resulting power-basis coefficients to the product Bernstein basis of degree $(3, 3)$. If

$$P(s, t) = \sum_{p,q=0}^{3} a_{pq} s^p t^q,$$

the coefficient of $B_i^3(s)B_j^3(t)$ is computed from the exact identity

$$b_{ij} = \sum_{p=0}^{i} \sum_{q=0}^{j} a_{pq} \frac{\binom{i}{p}}{\binom{3}{p}} \frac{\binom{j}{q}}{\binom{3}{q}}.$$

The script checks that the resulting matrix agrees entry by entry with Proposition A.2 and that all 16 coefficients are strictly positive.

3. Product lower bound for Example 4

This section documents Proposition A.3 using the ancillary source `certify_prop_A3_mpfr.cpp`; the supplied run is recorded in `certify_prop_A3_mpfr_output.txt`.

In complemented coordinates the only nontrivial domain for the right-hand side of the product inequality is

$$1/64 \le x \le y \le 63/64.$$

After independently verifying the bracket for x_* used in φ, the source refines the triangle by vertical and horizontal lines through the listed breakpoints. For $\widetilde{a}$ these breakpoints are $1/2$ and $1/\sqrt{2}$; for χ they are $1/64, 1/16, 15/16, 63/64$; for D they are $1/2$ and $181/256$. Each resulting cell uses the corresponding fixed formulas, except for the cell meeting the rounded approximation of $1/\sqrt{2}$ described below. Diagonal boxes are subdivided into two smaller diagonal triangles and the rectangle between them, preserving $x \le y$ throughout.

The exact breakpoint $1/\sqrt{2}$ of $\widetilde{a}$ is replaced by the binary64 value obtained by evaluating $\sqrt{1/2}$ with MPFR and rounding upward. Since this rounded value is not smaller than the exact breakpoint, the affected cell may extend slightly across $1/\sqrt{2}$. On that cell the interval for $1 - 2x^2$ is restricted to $[0, \infty)$ before the square root is evaluated, so the computation remains valid across the exact breakpoint and the approximation leaves no uncovered points.

On a terminal cell the source first attempts direct interval evaluation of

$$K_4(x, y) - \varphi(x)\varphi(y).$$

If this does not give a positive lower endpoint, it evaluates the function at the rounded midpoint and bounds its variation using interval bounds for both first derivatives and outward midpoint radii. The complete cover has

$$1399 \text{ terminal cells}, \qquad 1348 \text{ splits}, \qquad \text{maximum depth } 17,$$

with no unresolved cell. The least certified lower bound is

$$2.3311935604120031 \cdot 10^{-8}.$$

4. Scalar checks for the analytic regions

This section documents Proposition A.4 using the ancillary source `certify_prop_A4_mpfr.cpp`; the supplied run is recorded in `certify_prop_A4_mpfr_output.txt`.

The driver recomputes the constants needed in Proposition A.4 and obtains the m_{cert} interval by subtracting an outward interval for the exact rational number $41/2000000$.

With $c_\varepsilon = m_{\mathrm{cert}}(1-\theta)\alpha$, the source reports

$$c_\varepsilon \ge 0.53084624205541608.$$

The scalar lower bounds for the small-t region, the boundary-region check, and the large-t tail are, respectively,

$$0.035919173514281852, \qquad 6.4360829338080744 \cdot 10^{-6}, \qquad 0.024386055203189365.$$

5. Local Hessian and stationary point

This section documents Proposition A.5 using the ancillary source `certify_prop_A5_mpfr.cpp`; the supplied run is recorded in `certify_prop_A5_mpfr_output.txt`.

On the local rectangle

$$0.352 \le t \le 0.450, \qquad 0 \le d \le 0.005,$$

a 40×20 grid gives

$$\min(\mathcal{N})_{tt} = 0.22836139797078275,$$
$$\min(\mathcal{N})_{dd} = 0.68037764661274025,$$
$$\min \det D^2\mathcal{N} = 0.17291110546253904.$$

Directed derivative evaluations at $t = 0.35615880$ and $t = 0.35615895$ have opposite signs. Evaluation on the entire intervening interval gives

$$\mathcal{N}(t, 0) \ge 6.7220206068085965 \cdot 10^{-7}.$$

At the former equality point of the two-construction estimate the directed lower bound is

$$\mathcal{N}(t_*, 0) \ge 2.9270160742389521 \cdot 10^{-5}.$$

6. Finite cover of the remaining compact region

This section documents Proposition A.6 using the ancillary source `certify_prop_A6_mpfr.cpp`; the supplied run is recorded in `certify_prop_A6_mpfr_output.txt`.

The remaining compact domain is handled by the centered mean-value form in (t, ξ), with analytic removal of boxes adjacent to $v = t(1 - \xi) = 0$ and strict-convexity removal of boxes contained in the local rectangle. Before starting the finite cover, the A.6 driver repeats the boundary check at $t = 10^{-5}$ and $v = 10^{-7}$; since the analytic lower bound is increasing in t and decreasing in v, this covers $t \geq 10^{-5}$ and $0 \leq v \leq 10^{-7}$. The verifier handles the piecewise-C^1 case by splitting at the breakpoints and bounding the derivative on each resulting piece. The complete complementary cover has

4557 terminal cells, 4384 splits, 10 local cells, maximum depth 12,

and no unresolved cell. Its least lower bound is

$$5.3640462249763984 \cdot 10^{-9}.$$

Reproduction record

The six supplied output files are the complete standard-output streams of the six proposition-specific verification programs. Reproduction succeeds when the five C++ executables and the exact Python checker exit with status zero and end with their proposition-specific pass messages; the README gives the complete build commands. The package was tested from a clean extraction with MPFR 4.2.2, and the A.2 output was generated by the exact rational checker. Both LaTeX documents compiled without warnings or unresolved references, and all six ancillary checks passed.

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